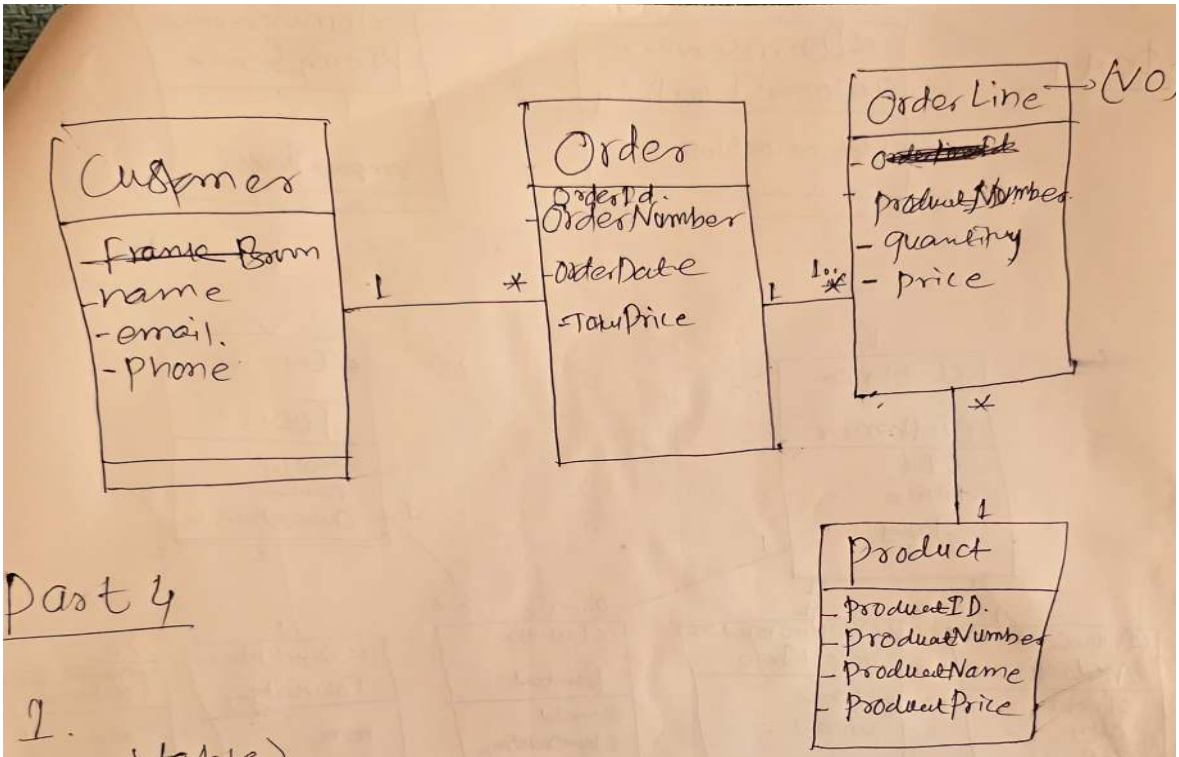
Lab3 Solution

Suppose you need to store the following order in the database:
Ordernumber:122435
Orderdate 11/09/2021
Customer name: Frank Brown
Customer email: fbrown@gmail.com
Customer phone: 0623156543
Total price : 5160.00

quantity	Product number	Product name	price
2	A546	IPhone 12	980.00
4	S333	Samsung Galaxy 12S	800.00

1. Draw the tables including data that you need to store this order in a relational database.



Part 4

1.

(table)

Customer

CustomerID	CustomerName	CustomerEmail	CustomerPhone
100	Frank Brown	fbrown@gmail.com	0623156543

Order

OrderNumber	Order Date	CustomerID	totalPrice
122435	11/09/2021	100	5160.00

OrderLine

OrderNumber	ProductNumber	ProductName	Quantity	UnitPrice
122435	A546	Iphone 12	2	980.00
122435	S333	Samsung Galaxy 12S	4	800.00

2. Draw the collections including data that you need to store this order in a mongo database.

2. MongoDB

Customers

```
{
  "_id" : "100",
  "name" : "Frank Boom",
  "email" : "fboom@gmail.com",
  "phone" : "0623156543"
}
```

Orders

```
{
  "_id" : 122435,
  "Order Date" : "11/09/2021",
  "totalPrice" : 5160.00,
  "customer" : {
      "customerId" : "100",
      "name" : "Frank Brown",
      "email" : "fbrown@gmail.com",
      "phone" : "0623156543"
    },
  "items" : [
      { "productNumber" : "A546", "productName" : "Iphone 12", "quantity" : 2,
        "unitPrice" : 980.00},
      { "productNumber" : "S333", "productName" : "Samsung Galaxy 12S",
        "quantity" : 4, "unitPrice" : 800.00}
    ]
}
```

3. Draw the tables including data that you need to store this order in a cassandra database if you are interested in orders by customer.

3. Cassandra

Orders_by_customer

customer_email	order_date	order_number	customer_name	customer_phone	total_price	product_number	product_name	quantity	unit_price
fbroom@gmail.com	11/09/2021	122435	Frank Broom	0623156543	5160.00	A546	Iphone 12	2	980.00
fbroom@gmail.com	11/09/2021	122435	Frank Broom	0623156543	5160.00	A333	Samsung Galaxy 12S	4	800.00

Back

4. Draw the database structure including data that you need to store this order in a neo4j database.

4. Neo4j graph

Nodes

- (:Customer {id: '100', name: 'Frank Brown', email: 'fbrown@gmail.com', phone: '0622156543'})

(:Order {orderNumber: 122435, orderDate: '11/03/2021', totalPrice: 5160.00})

(:Product {productNumber: 'A546', productName: 'Iphone 12', price: 980.00})

(:Product {productNumber: 'S333', productName: 'Samsung Galaxy 12S', price: 800.00})

Relationships

(Customer 100) - [:PLACED] -> (Order 122435)

(Order 122435) - [:CONTAINS {quantity: 2, unitPrice: 980.00}] -> (Product A546)

(Order 122435) - [:CONTAINS {quantity: 4, unitPrice: 800.00}] -> (Product S333)